

AN EQUIVARIANT K -THEORETIC MOLEV–SAGAN TYPE FORMULA FOR DOUBLE GROTHENDIECK POLYNOMIALS

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ABSTRACT. We describe the multiplication rule for double Grothendieck polynomials implemented in `schubmult.mult.groth.double`. It rests on a conjectural closed formula for the coefficients of $\prod_{i \leq k} (x_i + z) \mathfrak{G}_u(x; y)$ in the $\mathfrak{G}$ -basis: each coefficient is a factorial elementary symmetric polynomial $E_{n,n}(\mathcal{Y}; -z)$ in an alphabet $\mathcal{Y}$ built from the formal inverses $\ominus y_a$ of the fixed window values and one copy of $-1/\beta$ per left-moving value, times a prefactor of powers of β and atoms $(1 + \beta y_a)^{-1}$. At $\beta = 0$ this is the Pieri formula of [5, Theorem 7.1]. Because the coefficient is itself a factorial elementary symmetric polynomial, its z -divided differences are again of that form, which gives a closed Pieri rule for every $E_{p,k}(x; z)$ and, through the elementary-symmetric expansion of double Schubert polynomials, a Molev–Sagan type expansion of $\mathfrak{G}_u(x; y) \mathfrak{G}_v(x; z)$ in the basis $\{\mathfrak{G}_w(x; y)\}$. The formula has been verified exhaustively on S_4 and sampled through S_6 .

1. CONVENTIONS

Throughout, β is a formal parameter and we write

$$a \oplus b = a + b + \beta ab, \quad \ominus a = \frac{-a}{1 + \beta a},$$

for the multiplicative formal group law and its formal inverse, so that $a \oplus (\ominus a) = 0$ and $1 + \beta(a \oplus b) = (1 + \beta a)(1 + \beta b)$. Two identities are used repeatedly:

$$(1) \quad 1 + \beta(\ominus a) = \frac{1}{1 + \beta a}, \quad z \oplus (\ominus a) = \frac{z - a}{1 + \beta a}.$$

Permutations are written in one-line notation and $\ell(w)$ is the length (number of inversions). For $u \leq w$ in Bruhat order we write $\ell(u, w) = \ell(w) - \ell(u)$. The transposition exchanging positions $a < b$ is t_{ab} .

Double Schubert polynomials. $\mathfrak{S}_w(x; y)$ is the double Schubert polynomial in the convention of [5]: for $w_0 \in S_n$ the longest element, $\mathfrak{S}_{w_0}(x; y) = \prod_{i+j \leq n} (x_i - y_j)$, and $\mathfrak{S}_{ws_i} = \partial_i^x \mathfrak{S}_w$ when $\ell(ws_i) < \ell(w)$. Thus $\mathfrak{S}_w(x; y)$ vanishes at $x_i = y_{w(i)}$ for all i , i.e. at the torus fixed point indexed by w . This is the convention of `schubmult`'s `DSx`.

Double Grothendieck polynomials. $\mathfrak{G}_w(x; y)$ is the double β -Grothendieck polynomial [1] normalized so that

$$\mathfrak{G}_{w_0}(x; y) = \prod_{i+j \leq n} (x_i \oplus y_j), \quad \mathfrak{G}_{ws_i} = \pi_i^x \mathfrak{G}_w \text{ if } \ell(ws_i) < \ell(w),$$

where $\pi_i^x f = \partial_i^x ((1 + \beta x_{i+1})f)$ is the isobaric divided difference. This is the convention of `schubmult`'s `DGx`: for example

$$\mathfrak{G}_{21}(x; y) = x_1 \oplus y_1, \quad \mathfrak{G}_{312}(x; y) = (x_1 \oplus y_1)(x_1 \oplus y_2), \quad \mathfrak{G}_{231}(x; y) = (x_1 \oplus y_1)(x_2 \oplus y_1).$$

With this normalization $\mathfrak{G}_w(x; y)$ vanishes at the point $x_i = \ominus y_{w(i)}$, and the lowest-degree part of $\mathfrak{G}_w(x; y)$ is $\mathfrak{S}_w(x; -y)$. In particular

$$(2) \quad \mathfrak{G}_w(x; y) \Big|_{\beta=0} = \mathfrak{S}_w(x; -y),$$

so a formula for $\mathfrak{G}$ specializes to one for $\mathfrak{S}$ after $y \mapsto -y$.

Factorial elementary symmetric polynomials. For $0 \leq p \leq k$, $E_{p,k}(x; z)$ is the factorial elementary symmetric polynomial

$$E_{p,k}(x; z) = \sum_{1 \leq i_1 < \dots < i_p \leq k} \prod_{j=1}^p (x_{i_j} - z_{i_j-j+1}) = \mathfrak{S}_{c[p,k]}(x; z), \quad c[p, k] = s_{k-p+1} \cdots s_{k-1} s_k,$$

so $E_{0,k} = 1$ and $E_{k,k}(x; z) = \prod_{i=1}^k (x_i - z_1)$. In the variable $t = z_1$ the top one is a generating function for the ordinary elementary symmetric polynomials,

$$(3) \quad E_{k,k}(x; t) = \prod_{i=1}^k (x_i - t) = \sum_{p=0}^k (-t)^{k-p} e_p(x_1, \dots, x_k),$$

and the others are obtained from it by divided differences in z :

$$(4) \quad E_{p,k}(x; z) = (-1)^{k-p} \partial_{k-p}^z \cdots \partial_2^z \partial_1^z E_{k,k}(x; z_1), \quad \partial_j^z f = \frac{f - s_j^z f}{z_j - z_{j+1}}.$$

On monomials in $t = z_1$ the operator $\partial_q^z \cdots \partial_1^z$ acts by $t^m \mapsto h_{m-q}(z_1, \dots, z_{q+1})$, the complete homogeneous symmetric polynomial (zero when $m < q$).

2. THE CLASSICAL PIERI FORMULA

Sottile's k -Bruhat relation $u \xrightarrow{k} w$ holds when there are transpositions $t_{a_1 b_1}, \dots, t_{a_p b_p}$ with $a_i \leq k < b_i$, the lower indices a_i pairwise distinct, the upper indices $b_1 \geq b_2 \geq \dots \geq b_p$, each step a Bruhat cover, and $w = ut_{a_1 b_1} \cdots t_{a_p b_p}$; then $\ell(u, w) = p$. Write

$$P_k(u, w) = \{ u(i) : i \leq k, u(i) = w(i) \}$$

for the values in the window $\{1, \dots, k\}$ that are left in place.

Theorem 2.1 ([5, Theorem 7.1]). *For $u \in S_\infty$, $k \geq 1$ and $0 \leq p \leq k$,*

$$\mathfrak{S}_u(x; y) E_{p,k}(x; z) = \sum_{u \xrightarrow{k} w} E_{p-\ell(u,w), k-\ell(u,w)}(y_{P_k(u,w)}; z) \mathfrak{S}_w(x; y),$$

where $y_{P_k(u,w)}$ denotes the alphabet $\{y_a : a \in P_k(u, w)\}$ ($E_{p,k}$ is symmetric in its first alphabet, so the order is immaterial).

The case $p = k$ is a product formula: $E_{k-\ell, k-\ell}(y_P; z) = \prod_{a \in P} (y_a - z_1)$, i.e.

$$(5) \quad \mathfrak{S}_u(x; y) \prod_{i=1}^k (x_i - z) = \sum_{u \xrightarrow{k} w} \prod_{a \in P_k(u,w)} (y_a - z) \mathfrak{S}_w(x; y).$$

Each factor has a localization meaning: at a fixed window position i the polynomial $\mathfrak{S}_u$ “knows” $x_i = y_{u(i)}$, and the factor $(x_i - z)$ contributes its value there. Moved positions contribute 1, and $u \xrightarrow{k} w$ forces exactly one unit of length per moved position.

3. THE K -THEORETIC TOP-BLOCK FORMULA

Fix u , k , and w and look at the k window positions. Each $i \leq k$ falls into exactly one of three classes according to the fate of the value $u(i)$ in w :

$$\text{Fix}_k(u, w) = \{ i \leq k : w(i) = u(i) \},$$

$$\text{Left}_k(u, w) = \{ i \leq k : u(i) = w(j) \text{ for some } j < i, j \leq k \} \quad (\text{moves left inside the window}),$$

$$\text{Out}_k(u, w) = \{ i \leq k \} \setminus (\text{Fix}_k \cup \text{Left}_k) \quad (\text{leaves the window, or moves right inside it}).$$

Let $m_k(u, w) = k - |\text{Fix}_k(u, w)|$ be the number of moved window positions and $d = \ell(u, w)$. Attach to the pair (u, w) the *prefactor*

$$(6) \quad \rho_k(u, w) = \beta^{\ell(u,w)-m_k(u,w)} (-\beta)^{|\text{Left}_k(u,w)|} \prod_{i \in \text{Out}_k(u,w)} \frac{1}{1 + \beta y_{u(i)}},$$

and the alphabet of size $n_k(u, w) = |\text{Fix}_k(u, w)| + |\text{Left}_k(u, w)| = k - |\text{Out}_k(u, w)|$

$$(7) \quad \mathcal{Y}_k(u, w) = (\ominus y_{u(i)})_{i \in \text{Fix}_k(u, w)} \cup \left(-\frac{1}{\beta}\right)^{|\text{Left}_k(u, w)|},$$

the fixed values run through the formal inverse and each left-mover contributes the constant $-1/\beta$.

Definition 3.1 (The support). $\text{Supp}_k(u)$ is the set of w such that $\mathfrak{G}_w(x)$ occurs in $\prod_{i \leq k} (x_i + z) \mathfrak{G}_u(x)$, the product of *single* Grothendieck polynomials with z a formal variable. Equivalently, $\text{Supp}_k(u) = \bigcup_{p=0}^k \text{Supp}(e_p(x_1, \dots, x_k) \mathfrak{G}_u(x))$ which is the set of endpoints of the marked k -Pieri chains of Lenart–Sottile [3].

Conjecture 3.2 (Equivariant K -theoretic Pieri formula, top block). *For every $u \in S_\infty$, $k \geq 1$ and formal z ,*

$$(8) \quad \prod_{i=1}^k (x_i + z) \mathfrak{G}_u(x; y) = \sum_{w \in \text{Supp}_k(u)} \rho_k(u, w) E_{n, n}(\mathcal{Y}_k(u, w); -z) \mathfrak{G}_w(x; y), \quad n = n_k(u, w).$$

Since $E_{n, n}(\mathcal{Y}; -z) = \prod_{Y \in \mathcal{Y}} (Y + z)$, and $(-\beta)(-\frac{1}{\beta} + z) = 1 - \beta z$, the summand of (8) is the product

$$(9) \quad \beta^{\ell(u, w) - m_k(u, w)} \prod_{i \in \text{Fix}_k(u, w)} (z + \ominus y_{u(i)}) \prod_{i \in \text{Left}_k(u, w)} (1 - \beta z) \prod_{i \in \text{Out}_k(u, w)} \frac{1}{1 + \beta y_{u(i)}},$$

which is the form in which the rule is implemented. Using (1) the three kinds of factors can be written uniformly as

$$z + \ominus y_{u(i)} = \frac{z(1 + \beta y_{u(i)}) - y_{u(i)}}{1 + \beta y_{u(i)}}, \quad 1 - \beta z = 1 + \beta(-z), \quad \frac{1}{1 + \beta y_{u(i)}} = 1 + \beta(\ominus y_{u(i)}).$$

In particular every coefficient is a polynomial in β, z, y divided by a product of the atoms $1 + \beta y_a$, one for each moved-or-fixed window value except the left-movers; the β^{-1} in the alphabet is always cancelled by the $(-\beta)^{|\text{Left}|}$ in ρ_k .

Remark 3.3 (Why $-1/\beta$ and not 1). One is tempted to write the left-mover factor $1 - \beta z$ as “ $1 - (\beta z)$ ” and use the alphabet $(\ominus y_{\text{Fix}}, 1^{|\text{Left}|})$ against a second alphabet mixing $-z$ and βz . That product is not a factorial elementary symmetric polynomial in a single z -alphabet, and the divided differences of Section 4 would then have to be worked out term by term. Rescaling instead by $-\beta$ puts every factor in the shape $Y - (-z)$ with a common second variable, which is exactly what makes $E_{n, n}(\mathcal{Y}; -z)$ available and, with it, the closed form of Corollary 4.1.

Remark 3.4 (Consistency checks). (1) *Specialization $\beta = 0$.* In the product form (9) the powers $\beta^{\ell(u, w) - m_k(u, w)}$ kill every w with $\ell(u, w) > m_k(u, w)$; what survives are the w reached by exactly one Bruhat cover per moved window position, which are Sottile’s chains $u \xrightarrow{k} w$. The fixed factors become $z - y_{u(i)}$ and the others 1. Substituting $y \mapsto -y$ and $z \mapsto -z$ and using (2) gives (5), i.e. Theorem 2.1 with $p = k$.

(2) *The diagonal term.* Evaluating both sides of (8) at the fixed point $x_i = \ominus y_{u(i)}$ kills every $\mathfrak{G}_w$ with $w \neq u$ on the right (only $w \leq u$ survive localization at u , and $w \in \text{Supp}_k(u)$ forces $w \geq u$). The left side becomes $\prod_{i \leq k} (z + \ominus y_{u(i)}) \mathfrak{G}_u(\ominus y_u; y)$, which is the $w = u$ term of (8). So the coefficient of $\mathfrak{G}_u$ is correct unconditionally; it is the K -theoretic form of the localization reading of (5).

(3) *Non-equivariant limit.* At $y = 0$ the fixed factors are z , the out-factors are 1, and (8) reads $\prod (x_i + z) \mathfrak{G}_u = \sum_w \beta^{\ell(u, w) - m} z^{|\text{Fix}|} (1 - \beta z)^{|\text{Left}|} \mathfrak{G}_w$. Expanding in z recovers the non-equivariant Pieri coefficients of [3], and shows that the equivariant and non-equivariant supports coincide, which is why $\text{Supp}_k(u)$ could be defined non-equivariantly.

Remark 3.5 (On the support). The support is genuinely smaller than the set of endpoints of Bruhat chains $u = w_0 < w_1 < \dots < w_d = w$ by transpositions $t_{a_j b_j}$ with $a_j \leq k < b_j$ and weakly decreasing b_j (lower indices may now repeat). For instance with $u = 1243$ and $k = 2$ the permutation $w = 3412$ is such an endpoint, but $w \notin \text{Supp}_2(1243)$; the product formula would assign it the nonzero value $\beta/((1 + \beta y_1)(1 + \beta y_2))$, so the restriction to $\text{Supp}_k(u)$ in (8) cannot be dropped. The implementation enumerates the marked chains of [3] (`schub.lib.elem.sym.chains.groth`) to decide membership.

Example 3.6. $u = 21$, $k = 2$. The window values are $u(1) = 2$, $u(2) = 1$, and $\text{Supp}_2(21) = \{21, 231, 312, 321\}$:

w	Fix	Out	$\ell(u, w) - m$	coefficient of $\mathfrak{S}_w(x; y)$
21	$\{1, 2\}$	$\emptyset$	0	$(z + \ominus y_2)(z + \ominus y_1)$
231	$\{1\}$	$\{2\}$	0	$(z + \ominus y_2) \frac{1}{1 + \beta y_1}$
312	$\{2\}$	$\{1\}$	0	$(z + \ominus y_1) \frac{1}{1 + \beta y_2}$
321	$\emptyset$	$\{1, 2\}$	0	$\frac{1}{(1 + \beta y_1)(1 + \beta y_2)}$

Hence

$$(x_1 + z)(x_2 + z) \mathfrak{S}_{21}(x; y) = (z + \ominus y_1)(z + \ominus y_2) \mathfrak{S}_{21} + \frac{z + \ominus y_2}{1 + \beta y_1} \mathfrak{S}_{231} + \frac{z + \ominus y_1}{1 + \beta y_2} \mathfrak{S}_{312} + \frac{\mathfrak{S}_{321}}{(1 + \beta y_1)(1 + \beta y_2)}.$$

Example 3.7. $u = 1243$, $k = 2$ shows all three classes and a nontrivial β -power. Here $u(1) = 1$, $u(2) = 2$, $\ell(u) = 1$, and $\text{Supp}_2(1243)$ has seven elements:

w	Fix	Left	Out	$\ell(u, w) - m$	coefficient
1243	$\{1, 2\}$			0	$(z + \ominus y_1)(z + \ominus y_2)$
1342	$\{1\}$		$\{2\}$	0	$(z + \ominus y_1)/(1 + \beta y_2)$
1423	$\{1\}$		$\{2\}$	0	$(z + \ominus y_1)/(1 + \beta y_2)$
1432	$\{1\}$		$\{2\}$	1	$\beta(z + \ominus y_1)/(1 + \beta y_2)$
2341		$\{2\}$	$\{1\}$	0	$(1 - \beta z)/(1 + \beta y_1)$
2413		$\{2\}$	$\{1\}$	0	$(1 - \beta z)/(1 + \beta y_1)$
2431		$\{2\}$	$\{1\}$	1	$\beta(1 - \beta z)/(1 + \beta y_1)$

In the last three rows the value 2 has moved from position 2 to position 1, hence is a left-mover; the value 1 has left the window.

4. FROM THE TOP BLOCK TO EVERY $E_{p,k}$

Conjecture 3.2 is stated for a single formal variable z , but that is enough to determine the coefficients of every $E_{p,k}(x; z) \mathfrak{S}_u$, exactly as in the Molev–Sagan approach to factorial Schur functions [4]: by (4), $E_{p,k}(x; z)$ is obtained from $E_{k,k}(x; z_1)$ by divided differences in z , and those operators can be pushed onto the coefficients in (8). The point of writing the coefficient as $\rho_k(u, w) E_{n,n}(\mathcal{Y}; -z)$ is that its z -divided differences are again factorial elementary symmetric polynomials, by the very same identity (4) applied to the alphabet $\mathcal{Y}$ instead of x .

Corollary 4.1 (Pieri rule for $E_{p,k}$). *Assume Conjecture 3.2. For $0 \leq p \leq k$, $q = k - p$, and any alphabet $z_1, z_2, \dots$,*

$$E_{p,k}(x; z) \mathfrak{S}_u(x; y) = \sum_{w \in \text{Supp}_k(u)} \kappa_k(u, w; q; z) \mathfrak{S}_w(x; y), \quad \kappa_k(u, w; q; z) = \rho_k(u, w) E_{n-q, n}(\mathcal{Y}_k(u, w); z_1, \dots, z_{q+1}),$$

with $n = n_k(u, w)$ and $\rho_k, \mathcal{Y}_k$ as in (6), (7) ($E_{n-q, n} = 0$ when $q > n$).

Proof. Put $t = z_1$ in (8) (so $-z = -t$): $E_{k,k}(x; t) \mathfrak{S}_u = \sum_w \rho_k(u, w) E_{n,n}(\mathcal{Y}; t) \mathfrak{S}_w$, an identity of polynomials in t with coefficients in the field generated by y and β (the ρ_k and $\mathcal{Y}$ do not involve t). Apply the operator $(-1)^q \partial_q^z \cdots \partial_1^z$ to both sides. On the left it produces $E_{p,k}(x; z) \mathfrak{S}_u$ by (4); on the right it acts only on $E_{n,n}(\mathcal{Y}; z_1)$, where (4) with x replaced by $\mathcal{Y}$ gives $E_{n-q, n}(\mathcal{Y}; z_1, \dots, z_{q+1})$. Linear independence of the $\mathfrak{S}_w(x; y)$ finishes the proof. $\square$

Setting every $z_j = 0$ gives the ordinary elementary symmetric polynomials, since $E_{r,n}(\mathcal{Y}; 0) = e_r(\mathcal{Y})$.

Corollary 4.2 (Pieri rule for e_p). *Assume Conjecture 3.2. For $0 \leq p \leq k$,*

$$e_p(x_1, \dots, x_k) \mathfrak{S}_u(x; y) = \sum_{w \in \text{Supp}_k(u)} \rho_k(u, w) e_{p-|\text{Out}_k(u, w)|}(\mathcal{Y}_k(u, w)) \mathfrak{S}_w(x; y).$$

Remark 4.3 (Specialization to Theorem 2.1). At $\beta = 0$ only w with $\ell(u, w) = m_k(u, w)$ survive in ρ_k . Write $L = |\text{Left}_k(u, w)|$ and $P = P_k(u, w)$, so $n = |P| + L$ and $|P| = k - m$. In $(-\beta)^L E_{n-q, n}(\ominus y_P, (-1/\beta)^L; z)$ the only terms that survive as $\beta \rightarrow 0$ are those using all L copies of $-1/\beta$, and each such copy contributes $(-\beta)(-1/\beta - z_j) = 1 + \beta z_j \rightarrow 1$; what remains is $E_{n-q-L, n-L}(-y_P; z) = E_{p-m, k-m}(-y_P; z)$. Since $u \xrightarrow{k} w$ has $\ell(u, w) = m$, after $y \mapsto -y$ this is exactly the coefficient $E_{p-\ell, k-\ell}(y_P; z)$ of Theorem 2.1. Corollary 4.1 is thus the K -theoretic form of Theorem 2.1: the alphabet y_P of fixed values becomes $\ominus y_P$ padded by one $-1/\beta$ per left-mover, and the prefactor ρ_k carries the excess-length power of β and one atom $(1 + \beta y_a)^{-1}$ per value that leaves the window.

Example 4.4. Continuing Example 3.6 ($u = 21$, $k = 2$; no left-movers, so every alphabet consists of formal inverses):

w	$\rho_2(21, w)$	$\mathcal{Y}_2(21, w)$	n
21	1	$(\ominus y_2, \ominus y_1)$	2
231	$(1 + \beta y_1)^{-1}$	$(\ominus y_2)$	1
312	$(1 + \beta y_2)^{-1}$	$(\ominus y_1)$	1
321	$((1 + \beta y_1)(1 + \beta y_2))^{-1}$	$()$	0

Corollary 4.2 with $p = 1$ reads $e_1(\ominus y_1, \ominus y_2) \mathfrak{G}_{21} + e_0 \mathfrak{G}_{231} / (1 + \beta y_1) + e_0 \mathfrak{G}_{312} / (1 + \beta y_2)$ (321 has $p - |\text{Out}| = -1$ and drops out):

$$e_1(x_1, x_2) \mathfrak{G}_{21}(x; y) = (\ominus y_1 + \ominus y_2) \mathfrak{G}_{21} + \frac{\mathfrak{G}_{231}}{1 + \beta y_1} + \frac{\mathfrak{G}_{312}}{1 + \beta y_2},$$

while $p = 2$ gives $e_2(x_1, x_2) \mathfrak{G}_{21} = (\ominus y_1)(\ominus y_2) \mathfrak{G}_{21} + \frac{\ominus y_2}{1 + \beta y_1} \mathfrak{G}_{231} + \frac{\ominus y_1}{1 + \beta y_2} \mathfrak{G}_{312} + \frac{\mathfrak{G}_{321}}{(1 + \beta y_1)(1 + \beta y_2)}$. For $E_{1,2}(x; z) = x_1 + x_2 - z_1 - z_2$ ($q = 1$), Corollary 4.1 replaces $e_1(\ominus y_1, \ominus y_2)$ by $E_{1,2}(\ominus y_1, \ominus y_2; z_1, z_2) = \ominus y_1 + \ominus y_2 - z_1 - z_2$ and leaves the other two coefficients ($E_{0,1} = 1$) unchanged:

$$\mathfrak{G}_{21}(x; y) E_{1,2}(x; z) = (\ominus y_1 + \ominus y_2 - z_1 - z_2) \mathfrak{G}_{21} + \frac{\mathfrak{G}_{231}}{1 + \beta y_1} + \frac{\mathfrak{G}_{312}}{1 + \beta y_2}.$$

A left-mover appears in Example 3.7: for $w = 2341$ one has $\rho_2 = (-\beta)/(1 + \beta y_1)$, $\mathcal{Y}_2 = (-1/\beta)$, $n = 1$, so the $E_{2,2}(x; z)$ coefficient is $(-\beta)(-1/\beta - z_1)/(1 + \beta y_1) = (1 + \beta z_1)/(1 + \beta y_1)$ and the $E_{1,2}(x; z)$ coefficient is $(-\beta)E_{0,1}/(1 + \beta y_1) = -\beta/(1 + \beta y_1)$.

5. THE MOLEV-SAGAN TYPE FORMULA

The last ingredient is the expansion of a double Schubert polynomial into products of factorial elementary symmetric polynomials, which is what drives the double Schubert algorithm in `schubmult` [5]. For $v \in S_\infty$ let $\theta(v)$ be the code of the dominant permutation $\text{dom}(v)$ obtained by repeatedly sorting the code of v (Definition 5.1 below), and write $\theta_i = \theta_i(v^{-1})$. For permutations v', v'' and $i \geq 1$ write $v' \xrightarrow{i} v''$ if $v'' = v' t_{i, b_1} \cdots t_{i, b_r}$ with distinct $b_j \neq i$ and each step a Bruhat cover; set $\sigma_i(v', v'') = \#\{j : b_j < i\}$ and $D_i(v', v'') = \{v'(i)\} \cup \{v'(j) : v'(j) \neq v''(j)\}$.

Definition 5.1. μ is dominant if $\mathbf{c}_i(\mu) \geq \mathbf{c}_{i+1}(\mu)$ for all i . If v is not dominant let i be maximal with $\mathbf{c}_i(v) < \mathbf{c}_{i+1}(v)$ and set $\text{dom}(v) = \text{dom}(vs_i)$; otherwise $\text{dom}(v) = v$. Then $\theta_i(v) = \mathbf{c}_i(\text{dom}(v))$.

Theorem 5.2 ([5]). For $v \in S_\infty$, with r the number of nonzero parts of $\theta(v^{-1})$,

$$\mathfrak{S}_v(x; z) = \sum_{(v_0, \dots, v_r)} \prod_{i=1}^r (-1)^{\sigma_i(v_{i-1}, v_i)} E_{\theta_i - \ell(v_{i-1}, v_i), \theta_i}(x; z_{D_i(v_{i-1}, v_i)}),$$

the sum over chains $1 = v_0 \xrightarrow{1} v_1 \xrightarrow{2} \cdots \xrightarrow{r} v_r = v \text{ dom}(v^{-1})$, where z_D is the alphabet $\{z_a : a \in D\}$ in the order produced by `call_zvars`.

Each factor is an $E_{p,k}(x; \cdot)$ with $k = \theta_i$ and $p = \theta_i - \ell(v_{i-1}, v_i)$, so Corollary 4.1 can be applied layer by layer.

Theorem 5.3 (*K-theoretic Molev–Sagan formula, conditional on Conjecture 3.2*). For $u, v \in S_\infty$,

$$\mathfrak{S}_u(x; y) \mathfrak{S}_v(x; z) = \sum_w c_{u,v}^w(y; z) \mathfrak{S}_w(x; y), \quad c_{u,v}^w(y; z) = \sum_{\substack{(v_0, \dots, v_r) \\ (u_0, \dots, u_r)}} \prod_{i=1}^r (-1)^{\sigma_i(v_{i-1}, v_i)} \kappa_{\theta_i}(u_{i-1}, u_i; \ell(v_{i-1}, v_i); z_{D_i(v_{i-1}, v_i)}),$$

where the v -chain is as in Theorem 5.2, the u -chain has $u_0 = u$, $u_r = w$ and $u_i \in \text{Supp}_{\theta_i}(u_{i-1})$, and κ is the closed-form coefficient of Corollary 4.1,

$$\kappa_k(u', u''; q; z) = \rho_k(u', u'') E_{n_k(u', u'') - q, n_k(u', u'')}(\mathcal{Y}_k(u', u''); z_1, \dots, z_{q+1}).$$

Consequently, if $\mathfrak{S}_v(x; z) = \sum_{v'} a_{v,v'}(z; \beta) \mathfrak{S}_{v'}(x; z)$ is the (finite) expansion of $\mathfrak{S}_v$ into double Schubert polynomials, then

$$\mathfrak{S}_u(x; y) \mathfrak{S}_v(x; z) = \sum_w \left(\sum_{v'} a_{v,v'}(z; \beta) c_{u,v'}^w(y; z) \right) \mathfrak{S}_w(x; y).$$

Proof, given Conjecture 3.2. Substitute Theorem 5.2 for $\mathfrak{S}_v(x; z)$ and expand $\mathfrak{S}_u \cdot \prod_i E_i$ one factor at a time from $i = 1$ to r using Corollary 4.1, which is a formal consequence of (8) and (4). Each application replaces $\mathfrak{S}_{u_{i-1}}$ by $\sum_{u_i} \kappa_{\theta_i}(u_{i-1}, u_i; \dots) \mathfrak{S}_{u_i}$. The second statement is bilinearity. $\square$

At $\beta = 0$ every κ becomes the factorial elementary symmetric polynomial in the fixed values, the u -chains become chains of Sottile relations $u_{i-1} \xrightarrow{\theta_i} u_i$, and Theorem 5.3 is the Molev–Sagan type formula of [5] (after $y \mapsto -y$). The K -theoretic novelty is entirely in κ , and it is of the same shape as the classical coefficient: a factorial elementary symmetric polynomial, now in the alphabet $\ominus y_{\text{Fix}}$ padded with one $-1/\beta$ per left-mover, times the prefactor ρ carrying the excess-length power of β and the atoms $(1 + \beta y_a)^{-1}$ for values that exit or move right; and the u -steps are restricted to the marked-chain supports.

Example 5.4 (Headline example). $u = 21$, $v = 132$. The expansion of $\mathfrak{S}_{132}(x; z)$ into double Schubert polynomials is

$$\mathfrak{S}_{132}(x; z) = (1 + \beta z_1)^2 (1 + \beta z_2) \mathfrak{S}_{132}(x; z) + \beta (1 + \beta z_1) (1 + \beta z_2) \mathfrak{S}_{231}(x; z) + (z_1 \oplus z_2) (2 + \beta (z_1 \oplus z_2)),$$

and $\theta(132^{-1}) = \theta(231^{-1}) = (2)$, so both $\mathfrak{S}_{132}(x; z) = E_{1,2}(x; z)$ and $\mathfrak{S}_{231}(x; z) = E_{2,2}(x; z)$ are single layers with $k = 2$. Assembling them with the coefficient polynomials of Example 3.6 and simplifying,

$$\mathfrak{S}_{21}(x; y) \mathfrak{S}_{132}(x; z) = \gamma \mathfrak{S}_{21}(x; y) + (1 + \beta \gamma) (\mathfrak{S}_{231}(x; y) + \mathfrak{S}_{312}(x; y) + \beta \mathfrak{S}_{321}(x; y)),$$

$$\gamma = z_1 \oplus z_2 \oplus (\ominus y_1) \oplus (\ominus y_2) = \frac{(z_1 \oplus z_2) - (y_1 \oplus y_2)}{(1 + \beta y_1)(1 + \beta y_2)}, \quad 1 + \beta \gamma = \frac{(1 + \beta z_1)(1 + \beta z_2)}{(1 + \beta y_1)(1 + \beta y_2)}.$$

At $\beta = 0$ this is $\mathfrak{S}_{21}(x; -y) \mathfrak{S}_{132}(x; z) = (z_1 + z_2 - y_1 - y_2) \mathfrak{S}_{21}(x; -y) + \mathfrak{S}_{231}(x; -y) + \mathfrak{S}_{312}(x; -y)$, the familiar double Schubert product. The coefficient of $\mathfrak{S}_{21}$ is the formal-group-law sum of the same four terms, and the factor $1 + \beta \gamma$ multiplying the three higher terms is the “ K -theoretic 1”.

6. IMPLEMENTATION AND VERIFICATION

In `schubmult.mult.groth_double`:

- `grothmult_double_top` implements (8) (`_top_block_coeff` is the product of factors, `_top_block_support` is $\text{Supp}_k(u)$);
- `groth_elem_sym_frac` implements κ of Theorem 5.3 in the closed form of Corollary 4.1: it evaluates $E_{n-q,n}(\mathcal{Y}; z)$ by the subset expansion $\sum_{i_1 < \dots < i_p} \prod_j (Y_{i_j} - z_{i_j-j+1})$ as a dynamic program over the alphabet (7), clearing the denominator of each $\ominus y_a$ entry as it is used so that the numerator stays polynomial;
- `groth_schub_vpath_mul` implements $c_{u,v}^w$ by running the u -chains against the precomputed v -path dictionaries (`compute_vpathdicts`), exactly as `schubmult_double` does for double Schubert polynomials;
- `grothmult_double` implements the final displayed formula of Theorem 5.3, with `dgroth_to_dschub` supplying the coefficients $a_{v,v'}$.

Coefficients are carried throughout as “flat fractions” (a polynomial numerator together with exponents of the atoms $1 + \beta y_a$), which is possible precisely because (8) produces no other denominators. The ring interface is `DGx(u) * DGx(v, z)`, and `.simplify()` returns the coefficients in the factored form used above.

Conjecture 3.2 was checked against the exact fold `grothmult_double_block` (which multiplies by one linear factor $x_i + z$ at a time using the K_T -Chevalley formula of [2]) for every $u \in S_4$ and $1 \leq k \leq 3$, and for samples in S_5 and S_6 with $k \leq 5$; Theorem 5.3 was checked by expanding both sides as polynomials for every pair $u, v \in S_3$ and for samples of pairs in S_4 . All checks passed. A proof of Conjecture 3.2 is expected to follow from the localization characterization of the coefficients: the diagonal term is already forced, and the factors for moved positions should follow by induction on $\ell(u, w)$ from the K_T -Chevalley formula.

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